

Design Considerations of a GaN-based Three-Level Traction Inverter for Electric Vehicles

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Abstract—This paper demonstrates a GaN-based three-level inverter design for an electric vehicle (EV) traction application. The basic power block is a three-level active neutral point clamped (3L-ANPC) converter operating at 800 V DC bus voltage with the use of two paralleled 650 V, 60 A GaN E-HEMTs. The paper focuses on switching mode selection for better thermal performance of high-power 3L-ANPC inverters. The paper presents design details and guidelines for power layout structure and gate driver. The key parasitic inductances of the designed power block are extracted using ANSYS Q3D. A PLECS simulation model using the device loss parameters shows approximate power delivering capability of 41 kW for the designed forced air-cooled three-phase inverter. Experimental double pulse test results are presented with low voltage overshoot to verify the low inductance power loop design. Experimental results of the three-phase 3L-ANPC inverter operation at 70 kHz switching frequency are presented for delivered output power of 5.1 kW at 550 V DC.

Index Terms—GaN, Three-level Active Neutral Point Clamped (3L-ANPC), traction inverter, and electric vehicle.

I. INTRODUCTION

A high DC bus voltage (V_{DC}) of 800 V will benefit the EV fast chargers and also enable the design of faster electric motors with lower copper loss [1]. The use of traditional two-level inverters for a higher DC bus voltage will result in higher losses with the use of 1200 V rated switching devices [2]. Moreover, the high voltage dv/dt will cause more stress across the turn-turn and coil-coil insulation of the motor. This will require expensive insulation solutions for traction motors [3]. Typical approaches for dv/dt mitigation at inverter output include the use of LCR filters which add to the overall loss and result in a significant inductor voltage drop [4]. A simpler solution by slower switching of power devices for dv/dt stress mitigation leads to under-utilization of wide bandgap devices [5]. This has led to increased interest in three-level converter topologies, which can be realized with 600 V to 650 V rated power devices. The lower voltage switching enables higher efficiency, lower electromagnetic interference (EMI), and lesser motor loss [6]. Further, the reduced turn-turn voltage stress enables high dv/dt operation with no additional LCR filter. This paper selects a three-level active neutral point clamped (3L-ANPC) converter topology as the basic power design block. The 3L-ANPC topology offers the advantage of an active selection of zero

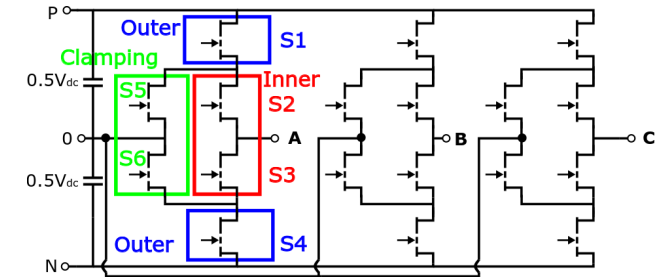


Fig. 1: Three phase 3L-ANPC topology showing the different switches: inner, outer and clamping.

states, allowing for smaller commutation loop inductance and better loss distribution compared to 3L-diode neutral point clamped (3L-DNPC) topology [7]. Also, 3L-ANPC offers better fault-tolerant operation for a single device short/open failure condition for a dual battery fed traction inverter [8].

Enhancement mode GaN HEMTs (GaN E-HEMTs) with lower switching charge (Q_g), zero reverse recovery loss, and smaller output capacitance produce significantly lower switching losses compared to similar rated SiC and Si power devices. A datasheet-based comparative evaluation chart of commercially available power semiconductor devices with different available ratings from various manufacturers is presented in Fig.2. A 650 V, 60 A GaN E-HMT (GS66516B) has been used in this paper for the design of the power block. GaN E-HMT manufacturers typically chose Si substrate for cost-benefit even though Si has significantly lower thermal conductivity compared to SiC ($k_{Si} = 1.5 \text{ W/cm K}$ and $k_{SiC} = 4.9 \text{ W/cm K}$) [9]. This leads to the degradation in efficiency of the GaN E-HEMTs at higher temperatures, particularly due to significantly high conduction losses. The temperature-dependent switching loss and on-state resistance ($R_{ds,on}$) characteristics of the selected device are shown in Fig.5a and Fig.5b respectively. Hence, careful attention needs to be given to the thermal design of the power block for high-power inverters. In this regard, this paper presents the preferred switching mode, power layout, and gate drive design approach.

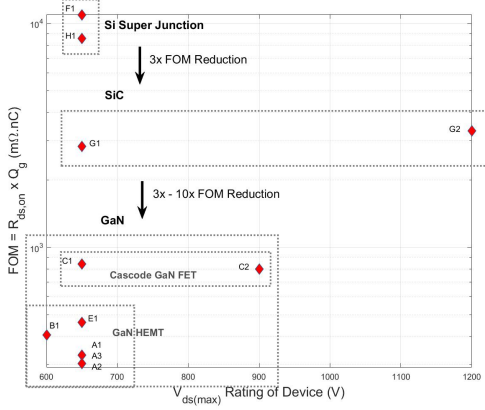


Fig. 2: FOM comparison of commercially available power semiconductor devices.

II. 3L-ANPC POWER BLOCK DESIGN

A. Switching Mode Selection

3L-ANPC topology enables several possible switching modes of operation to realize the P, 0, and N states. The P, 0 and N states correspond to the switching pole voltages (V_{AO}) $+0.5V_{dc}$, 0 and $-0.5V_{dc}$ respectively as shown in Fig.1. However, there are several switching possibilities to realize them. The use of the clamping switch produces two switching possibilities each for P and N states termed as ($P_C, P_{C'}$) and ($N_C, N_{C'}$). There are three switching possibilities for zero states termed as 0_+ , 0_- , and 0_F . The switching states are shown in Table.I.

TABLE I: Switching States of P, 0 and N states

	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆
P _C	1	1	0	0	0	1
P _{C'}	1	1	0	0	0	0
0 ₊	0	1	0	0	1	1
0 _F	0	1	1	0	1	1
0 ₋	0	0	1	0	1	1
N _{C'}	0	0	1	1	0	0
N _C	0	0	1	1	1	0

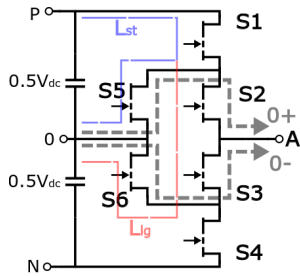
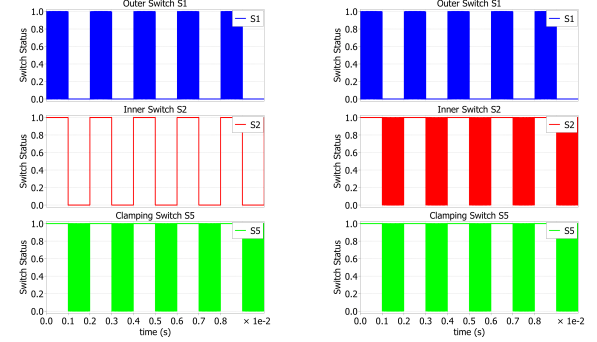


Fig. 3: Realization of different switching states in 3L-ANPC

The switching mode of 3L-ANPC influences the selection of devices and planning of power layout structure. Therefore it is considered the first step of the design process. It is preferred to select P_C and N_C for realizing P and N states to ensure that

the steady-state voltages across the off devices are clamped to half DC bus. It is also preferred to avoid the power loop commutation at full load current using the higher inductance path, i.e., the longer loop L_{lg} shown in Fig.3. Therefore the two switching modes of operation compared here for power block design are - (1) short-loop and (2) full mode switching [10].



(a) Short-loop switching mode.

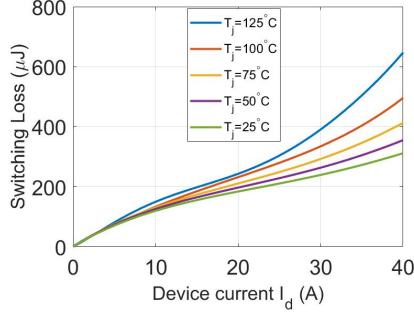
(b) Full switching mode

Fig. 4: High frequency and line frequency switching of outer, clamping and inner switches in 3L-ANPC for short-loop and full switching mode.

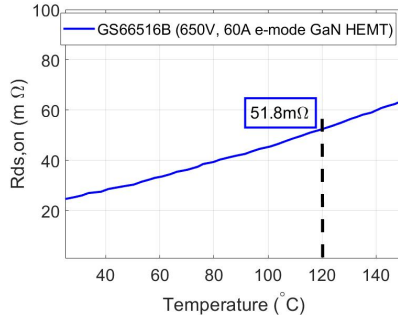
Switching between $P_C \leftrightarrow 0_+$ and $N_C \leftrightarrow 0_-$ states results in power loop commutation through short loop path only with lower inductance L_{st} shown in Fig.3. This is termed short-loop switching. Outer and clamping switches operate at switching frequency while the inner device operates at line frequency for this switching mode, as shown in Fig.4. Therefore to lower the overall system cost, Si-based MOSFETs are often selected over wide bandgap (WBG) devices for inner devices in hybrid 3L-ANPC converter designs [11]. However, the inner devices that conduct during both on and off state result in significantly high conduction losses. Moreover, the switching implementation of the short loop requires an additional decision logic to select between $0_+/0_-$ based on the previous state P_C/N_C .

For high-power GaN-based 3L-ANPC design, the conduction loss dominates the overall inverter loss. The inner and clamping device conduction losses are significantly reduced using full mode switching [10]. The 0_+ and 0_- paths shown in Fig.3 are operated in parallel for 0_F mode. Hence it reduces the zero state conduction loss to half. However, full mode switching operation requires all the devices to operate at switching frequency, as shown in Fig.4. Therefore GaN E-HEMTs are used for all the switches in 3L-ANPC operation in this work. The clamping and inner devices operate synchronously in parallel for the full mode, sharing half current. This keeps the overall switching losses in a full mode similar to the short-loop mode. The total loss is reduced with better distribution across the devices. Further, the switching logic implementation is simplified because of a single zero state 0_F . The top inner switch S_2 and bottom clamping switch S_6

share a common gate signal. Analogously, S_3 and S_5 share a common gate signal.



(a) Switching loss ($E_{on} + E_{off}$) vs junction temperature (Gate drive resistance: $R_{on}/R_{off} = 12\ \Omega / 4\ \Omega$)



(b) $R_{ds,on}$ vs junction temperature

Fig. 5: Switching and conduction loss variation of selected GaN device (GS66516B) with junction temperature.

A comparative evaluation of short-loop and full switching mode is done using PLECS thermal simulation. The simulation requires a thermal model of the packaged device with temperature dependent turn-on loss (E_{on}), turn-off loss (E_{off}) and on-state resistance $R_{ds,on}$. Towards that, the junction-case Caue RC network model of the packaged GaN E-HEMT is obtained from the device manufacturer datasheet [12]. For determination of E_{on} and E_{off} , the LTSpice model provided by the device manufacturer is used. The parasitic inductances input to the LTSpice simulation are extracted using ANSYS Q3D for the designed 3L-ANPC power block. The obtained switching losses and $R_{ds,on}$ of the selected device are shown in Fig.5a and Fig.5b respectively. The details of the case-heat-sink thermal design are discussed in Section.II-B. The electrical and thermal parameters used for the simulation are tabulated in Table.II and Table.III respectively.

TABLE II: Electrical Parameters selected for simulation

Parameter	Value	Description
V_{dc}	800 V	DC bus voltage
I_{pk}	30 A	Peak line-line current
m	0.3 - 1	Modulation index
pf	0.5 - 1	Power factor
f_{ref}	1 kHz	Fundamental frequency of 3L-ANPC
f_{sw}	50 kHz	Switching frequency

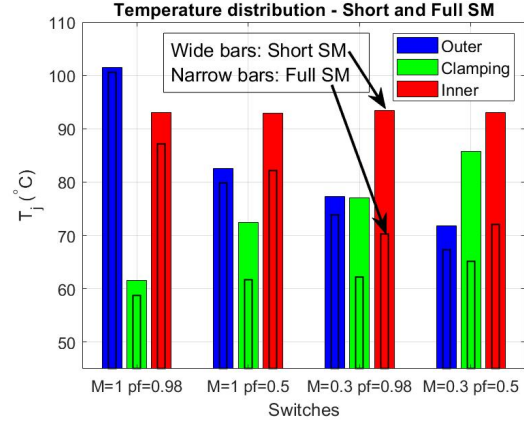


Fig. 6: Temperature rise of outer, clamping and inner switches during 3L-ANPC operation with short and full switching modes.

The thermal stress distribution across outer, inner, and clamping switches strongly depends on the operating modulation index (m) and power factor (pf). As we move to low m operation, the increased zero state period results in high conduction loss for inner and clamping switches. Similarly, as we move to low pf operation, the clamping switches get more often hard-switched, resulting in higher switching losses across them. The inner devices conduct during both P/N and 0 states. This makes inner device conduction losses always high irrespective of m and pf . These trends of thermal stress variation are observed in junction temperature (T_j) for outer, inner, and clamping devices in Fig.6. The thermal stress is determined for short-loop and full switching mode operation considering an ambient temperature of $T_{amb} = 45^\circ\text{C}$ and a peak current of 30 A through each GaN E-HEMT. Fig.6 shows a major reduction in T_j of inner and clamping switches achieved particularly at lower m using the full switching mode. Therefore under the worst case operating condition of $m = 1$ and unity power factor, the three-phase inverter rating is estimated at $P_{max} = 41\text{ kW}$. This estimation is for $T_{j(max)} = 125^\circ\text{C}$ considering $T_{amb} = 45^\circ\text{C}$ with two parallel GaN E-HEMTs delivering 34 A peak current per device at $V_{dc} = 800\text{ V}$.

B. Thermal Design

The previous section highlighted the selection of switching mode that enables lower heating of GaN E-HEMTs for higher efficiency. This section focuses on the thermal design approaches for a GaN-based high-power converter design. Using top-cooled devices with top-side heat sink mounting typically prevents the placement of key surface mount technology (SMT) components like decoupling capacitors and gate drive circuits close to the device. This problem is often resolved by the design of multilayer FR4 PCBs. Multilayer PCBs require thermal vias or pedestal heat sinks as cooling solution. A custom pedestal heat sink increases the design complexity and is not discussed in this work. Multilayer PCBs are often

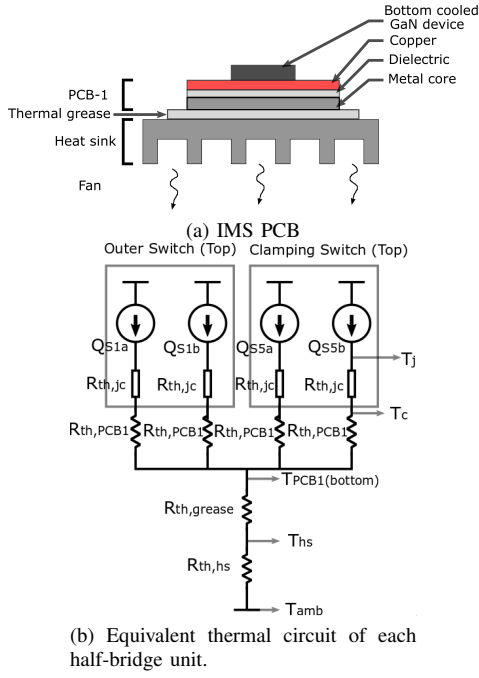


Fig. 7: Thermal network for the IMS PCB based 3L-ANPC. S_{1a} and S_{1b} represent the the two parallel GaN E-HEMTs which form the S_1 switch.

used to create vertical flux cancellation paths to enable ultra-low power loop inductance [13]. However, the thermal vias needed for heat dissipation in multilayer PCBs result in high case to heat sink thermal resistance ($R_{th(c-hs)}$). Hence such design approaches are limited to lower power applications. The presented design approach uses bottom-cooled devices with insulated metal core substrate (IMS) PCBs (PCB-1 shown in Fig.10), which offer low $R_{th(c-hs)}$ for GaN-based high power block design [14]. This limits the power layout design to a single-layer PCB, as shown in Fig.10a. The different layers involved in heat extraction from the bottom-cooled GaN E-HEMT are shown in Fig.7a. The resulting equivalent thermal circuit is shown in Fig.7b, and resistances are presented in Table.III. The thermal conductivity (k), dielectric constant (ϵ_r), and thickness of the insulation material in IMS are the key parameters of selection. The presented design considers a low-cost di-electric material with $k = 3 \text{ W/mK}$, $\epsilon_r = 4.8$ and thickness $t = 125 \mu\text{m}$. This results in estimated PCB thermal resistance of $R_{th,PCB1} = 2.92^\circ\text{C/W}$. The estimated device-to-ground plane stray capacitance is $C_p = 34 \text{ pF}$.

TABLE III: Thermal Parameters selected for simulation

Parameter	Value	Description
$R_{th,jc}$	0.27°C/W	Junction-case thermal resistance
$R_{th,PCB1}$	2.92°C/W	IMS PCB thermal resistance
$R_{th,grease}$	0.1°C/W	Interface paste thermal resistance
$R_{th,hs}$	0.3°C/W	Heat sink thermal resistance
T_{amb}	45°C	Ambient temperature

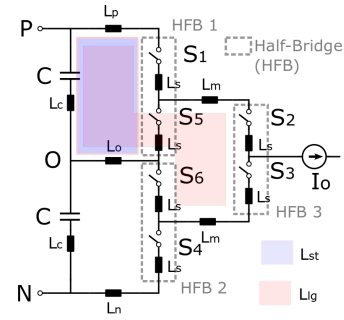
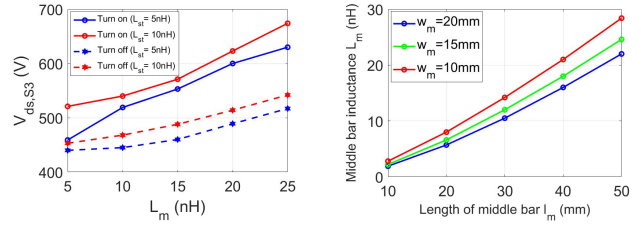


Fig. 8: Parasitic inductances in 3L-ANPC power block



(a) Maximum inner device (S_3) voltage during $0 \rightarrow P$ switching transient for full mode switching with short loop inductance (L_{st}) and middle bus bar inductance (L_m). (b) Design margin for deciding length and width of the middle bus bar.

Fig. 9: Impact of parasitic inductances on design of 3L-ANPC inverter.

C. Power Loop Design

The fast commutation of switches in a GaN-based 3L-ANPC requires careful attention to parasitic inductances to avoid device failure due to overvoltage. The key parasitic inductances in the 3L-ANPC are shown in Fig.8. The P, N and O bus bar inductances are denoted as L_p , L_n and L_o respectively. The source inductance is represented as L_s and the middle bus bar inductance is shown as L_m . The power loop design shown in Fig.8 is designed with an objective to realize a minimum short loop inductance L_{st} . ANSYS Q3D simulation is used to determine the estimated $L_{st} = L_c + L_p + L_o + 2L_s = 6.3 \text{ nH}$. L_{st} forms the inductance in the commutation path between outer and clamping switches. The 3L-ANPC power block is realized as three half-bridge power blocks HFB-1, HFB-2 and HFB-3 shown in Fig.8. The required vertical flux cancellation path for enabling lower L_{st} is realized on the top side through PCB-2 by use of low profile headers, as shown in Fig.10a. PCB-2 contains high voltage decoupling capacitors with high self-resonant frequency ($> 10 \text{ MHz}$) which serve as nearest charge reservoirs during switching transition to minimize overshoot. It should be noted that the provision for device current probing in the prototype design further adds to the power loop inductance.

The previous discussion has shown that full switching mode is preferred for high-power 3L-ANPC. In case of full mode the short loop (L_{st}) and long loop (L_{lg}) operate in parallel resulting in multi-loop commutation [15]. This can result in

a high voltage overshoot at S_3 during hard-switched turn-on of S_1 for $0 \rightarrow P$ operation in full mode. However, turning off of clamping switch minimizes the overvoltage [15] during the more severe turn-on transient. During the turn-off operation, the switches commutate half of the total current and therefore have a lower overshoot despite a higher inductance. This modified form of full-mode operation is used in the LTSPice simulation results used for analysis. A detailed discussion of the modified full mode operation is not considered under the scope of this paper. The maximum voltage seen across switch S_3 is determined for the two critical hard-switched cases in $0 \rightarrow P$ operation - (1) turn-on transient for switch S_1 and (2) synchronous turn-off switching operation of $S_5 - S_3$. The modified full mode switching operation is used and the results are presented in Fig.9. The allowable L_m for a given design L_{st} can be determined based on acceptable overvoltage stress. A single layer IMS PCB does not allow the flux cancellation path design for L_m . Therefore the length and width of the PCB trace need to be carefully selected for the L_m path. Fig.9b provides a look-up reference for deciding the length and width of the PCB trace to ensure that the design margin for L_m is ensured. Inductance is determined from the PCB trace dimensions for a selected 4oz copper thickness. For the designed, 3L-ANPC power block, Q3D simulation results estimate $L_m = 21$ nH.

D. Gate Drive Design

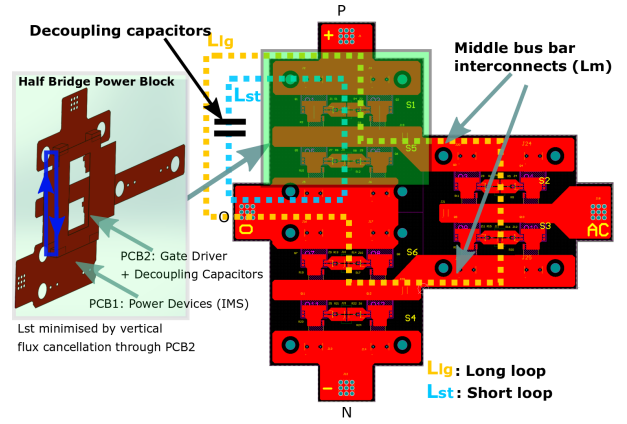
The complete gate drive consists of a totem pole stage and power supply stage formed by PCB stack assembly of PCB-2 and PCB-3 as shown in Fig.11. PCB-2 ensures a short and symmetrical gate drive loop for the two parallel connected GaN devices. Distributed gate and source resistors $R_{g,S1a} = R_{g,S1b} = 1 \Omega$ are used to avoid gate ringing between parallel devices. Commercially available gate driver ICs with $I_{source} = 3.6$ A and $I_{sink} = 8$ A are selected to enable fast switching of two parallel GaN devices. The bipolar gate drive bias voltage of 6 V and -4 V is generated using isolated DC-DC converters with very low $C_{iso} = 4$ pF. The gate discharge resistance and the anti-series connected zener diodes for protection are placed very close to the gate-source pads of the GaN E-HEMT. This ensures a safe gate-source voltage during switching transients.

III. EXPERIMENTAL RESULTS

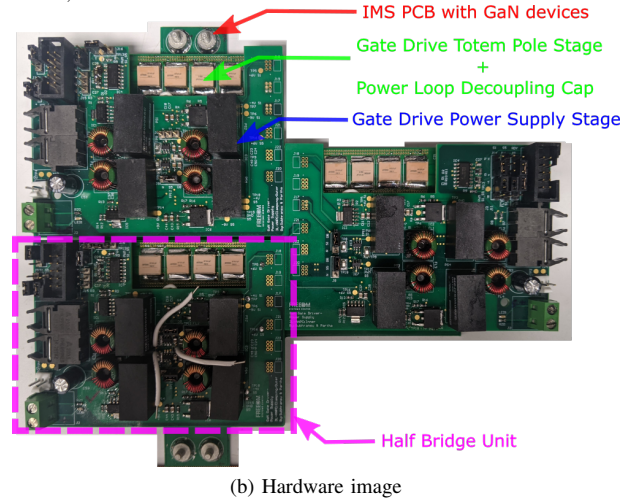
The designed GaN-based 3L-ANPC phase leg is shown in Fig.10. The complete gate drive stage with the decoupling capacitors is stacked on top of the power layout stage. The switching performance is evaluated using a double pulse test for the full switching mode. Preliminary experimental results of continuous three-phase inverter operation are also presented.

A. Double Pulse Test

The double pulse test (DPT) results with a single GaN E-HEMT per switch for full switching mode operation are shown in Fig.12. The gate resistances selected for the DPT are $R_{on} = 10 \Omega$, $R_{off} = 2 \Omega$, and $R_{g,S1a} = R_{s,S1a} =$



(a) Power layout structure of 3L-ANPC power block with two parallel 650 V, 60 A GaN E-HEMTs.



(b) Hardware image

Fig. 10: 3L-ANPC phase leg with integrated gate drive stage.

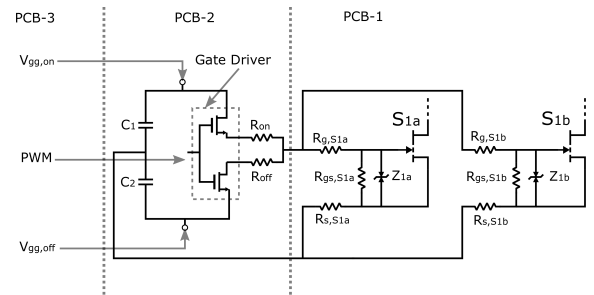
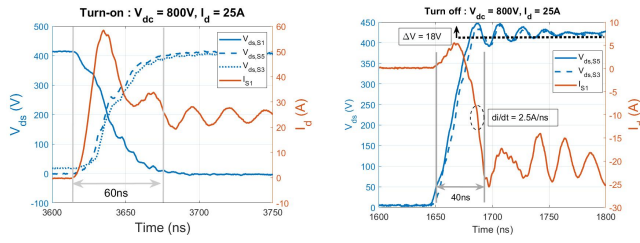


Fig. 11: Gate driver circuit.

1Ω . The test results show low overvoltage during turn-on and turn-off switching transient. The maximum voltage seen at S_3 during hard-switched turn-on of S_1 is $V_{ds,S3(max)} = 415$ V. The maximum voltage seen at S_3 during the synchronous turn-off of $S_3 - S_5$ is $V_{ds,S3(max)} = 446$ V. The turn-on switching loss of S_1 is $310 \mu\text{J}$ for $di/dt = 4$ A/ns and $dv/dt = 12$ V/ns at $V_{dc} = 800$ V and device current $I_d = 25$ A. Switches S_3 and S_5 operate synchronously during turn-off with a turn-off loss of $21 \mu\text{J}$ across each device.



(a) Hard-switched turn-on of S_1 during $0 \rightarrow P$ transition. This results in overvoltage at S_3 and S_5 due to multi-loop commutation. (b) Hard-switched turn-off of S_3 and S_5 during $0 \rightarrow P$ transition.

Fig. 12: Double pulse test (DPT) result for hard-switched $0 \rightarrow P$ operation with full switching mode operation.

The experimental results show noise-free safe switching operation of GaN E-HEMTs for 3L-ANPC. The short loop inductance can be determined from the turn-off switching transient of S_5 as $L_{st} \approx \Delta V / (di/dt) = 7.2 \text{ nH}$. This verifies the low short loop inductance of the design. The placement of voltage probing points close to drain-source terminals of GaN E-HEMTs is one of the major hindrances in capturing the actual device overvoltage waveforms.

B. Three-Level Inverter Test with RL load

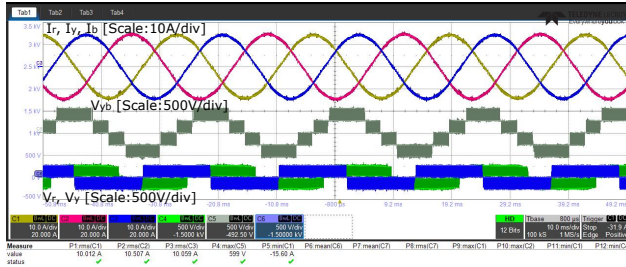


Fig. 13: 3-phase 3L-ANPC test results with RL load ($R = 16.4 \Omega$ and $L = 1 \text{ mH}$).

The experimental results of the three-phase 3L-ANPC inverter operation are shown in Fig.13. The results demonstrate a continuous operation of the inverter at switching frequency $f_{sw} = 70 \text{ kHz}$ delivering power $P_o = 5.1 \text{ kW}$. The DC bus voltage is at $V_{dc} = 550 \text{ V}$, and the fundamental frequency of the generated AC waveform is $f_{ref} = 60 \text{ Hz}$.

IV. CONCLUSION

The key design features of a GaN-based 3L inverter for EV traction application are presented for 800 V DC bus voltage. The simulation models estimate the power-delivering ability of 41 kW for the designed inverter with two parallel GaN E-HEMTs. Switching mode selection is key in lowering the junction temperature rise in GaN devices for 3L-ANPC operation. An IMS-based power layout design of the 3L-ANPC phase leg further improves the thermal performance to enable high efficiency from GaN E-HEMTs. The design guidelines and the effect of key parasitic inductances in the power layout structure

are highlighted. The double pulse test results of the designed 3L-ANPC phase leg demonstrate low voltage overshoots at 800 V DC bus and device current of 25 A. Preliminary three-phase inverter operation is demonstrated at 5.1 kW for 70 kHz switching frequency at 550 V DC.

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